# Basic SQL Queries and Aggregate Functions

Filtering, sorting, grouping, HAVING, and five aggregate functions

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**Project: Hotel Management System. Output convention: result tables are expected outputs from the fixed rows in the shared SQL script; they are not screenshots of MySQL execution.**

## 1. Filtering with WHERE

This query returns rooms currently marked AVAILABLE and sorts the output by room number.

```
SELECT room_id, room_number, room_status
FROM rooms
WHERE room_status = 'AVAILABLE'
ORDER BY room_number;
```

| room_id | room_number | room_status |
|---|---|---|
| 4 | 102 | AVAILABLE |
| 3 | 301 | AVAILABLE |
| 6 | 401 | AVAILABLE |
| 8 | 501 | AVAILABLE |

The room_number column is sorted as text. The expected result includes rooms 102, 301, 401, and 501.

## 2. Sorting with ORDER BY

```
SELECT type_name, base_rate
FROM room_types
ORDER BY base_rate DESC, type_name;
```

| type_name | base_rate |
|---|---|
| Suite | 7000.00 |
| Family | 5500.00 |
| Deluxe | 4000.00 |
| Standard | 2500.00 |

## 3. Aggregate functions

COUNT, SUM, AVG, MIN, and MAX summarize the four room types.

```
SELECT COUNT(*) AS room_type_count,
       SUM(base_rate) AS total_base_rate,
       AVG(base_rate) AS average_base_rate,
       MIN(base_rate) AS lowest_rate,
       MAX(base_rate) AS highest_rate
FROM room_types;
```

| room_type_count | total_base_rate | average_base_rate | lowest_rate | highest_rate |
|---|---|---|---|---|
| 4 | 19000.00 | 4750.000000 | 2500.00 | 7000.00 |

## 4. GROUP BY and HAVING

The first query groups rooms by hotel. The second query keeps room types that have at least two rooms.

```
SELECT h.city, COUNT(r.room_id) AS room_count
FROM hotels h JOIN rooms r ON r.hotel_id=h.hotel_id
```

```
GROUP BY h.hotel_id, h.city
ORDER BY h.city;

SELECT rt.type_name, COUNT(r.room_id) AS room_count
FROM room_types rt JOIN rooms r ON r.room_type_id=rt.room_type_id
GROUP BY rt.room_type_id, rt.type_name
HAVING COUNT(r.room_id) >= 2
ORDER BY rt.type_name;
```

| city | room_count |
|---|---|
| Bengaluru | 4 |
| Chennai | 4 |
| type_name | room_count |
| Deluxe | 2 |
| Standard | 3 |
| Suite | 2 |

The HAVING result excludes Family because only one room uses that type.

## 5. Screenshot guidance

Run each query separately in MySQL Workbench and capture the SQL statement with its Result Grid. Use the fixed sample data from Assignment 7 so that the values match these expected outputs.